Reports as Measurements

Matthias Michel's consciousness science, applied to LLM introspection

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Measuring AI Self-Reports

Overview

Part 1: The problem

1. Did you see it?
2. Two steps from answer to claim
3. The same problem in an LLM

Part 2: The right quantity?

4. A bad thermometer vs. the wrong instrument
5. The circle
6. Agreement can have a common cause

Part 3: Calibration

7. Check it against a different instrument
8. Model how the instrument could go wrong
9. What averaging fixes, and what it cannot

Part 4: Saying yes

10. The criterion problem
11. Signal detection in one picture
12. Injection makes the model say yes
13. How much of what it knows reaches the report?
14. Credit for noticing something

Part 5: Drift and limits

15. Validity drift
16. Where the analogy stops
17. Summary
18. Reading

Part 1

The problem

Did you see it?

A consciousness experiment, in its simplest form:

1. A word is flashed on a screen for 30 milliseconds, then covered by a pattern (a *mask*).
2. The subject is asked: *did you see a word?*
3. Repeat a few hundred times, with and without a word.

The researcher wants to know whether the subject was **conscious** of the word. The only thing the researcher gets is the **answer**.

So every conclusion about consciousness passes through a decision about what an answer means. Matthias Michel's work has studied this decision using the tools of measurement theory, drawing on Eran Tal and Hasok Chang.

Two steps from answer to claim

Michel splits the path from experience to conclusion into two steps:



In measurement language, the report is an **indication**: what the instrument shows, like the height of mercury in a thermometer. The claim is the **measurement outcome**, and getting from one to the other needs a model of the instrument.

Most worries about self-report are about Step 1 (is the subject accurate?). Michel's main worry is Step 2: should the answer be read as a reading of consciousness at all?

The same problem in an LLM

Lindsey (2025) runs the LLM version of the flash experiment:

1. Take the model's internal activation pattern for a word, such as *bread*.
2. Add it to the model's activations while it answers a question (*concept injection*).
3. Ask: *Do you detect an injected thought? If so, what is it about?*

Consciousness experiment

flashed word → experience → "I saw it"

Introspection experiment

injected vector → internal state → "I detect a thought about bread"

Michel writes about people and animals, not language models. The structure carries over anyway: the researcher changes something hidden, gets an answer, and has to decide what the answer measures.

Part 2

Is it measuring the right thing?

A bad thermometer vs. the wrong instrument

Michel's example (2023, p. 16):

Inaccurate

A thermometer that always reads 14 °C too high.
It measures temperature, badly.

Invalid

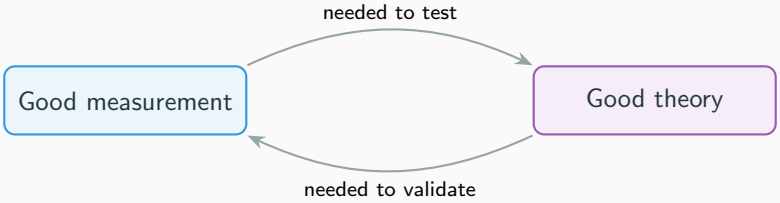
Using a thermometer to measure the GDP of Malawi.
It does not measure GDP at all.

Accuracy asks how close the reading is. **Validity** asks whether the reading is about the right quantity.

For introspection: a model that reports injected thoughts 20% of the time might be an inaccurate introspector, or its reports might track something else, such as how strange its own output looks.

The circle

To show that reports measure consciousness, you need a theory of consciousness. To test the theory, you need a measure.

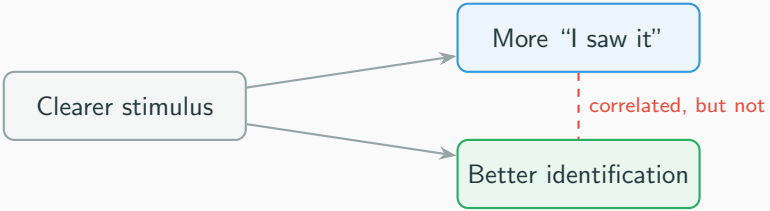


Michel calls this the **coordination problem**. Thermometry had the same circle, and Hasok Chang (*Inventing Temperature*, 2004) describes how it got out by iteration.

For LLMs: checking a report against the injected vector assumes the vector *is* the state being introspected.

Agreement can have a common cause

A natural check: people who say "I saw it" also identify the word better. Doesn't that show the reports are valid?



Michel's point (2019): clearer stimuli improve both, so the correlation does not need consciousness to explain it.

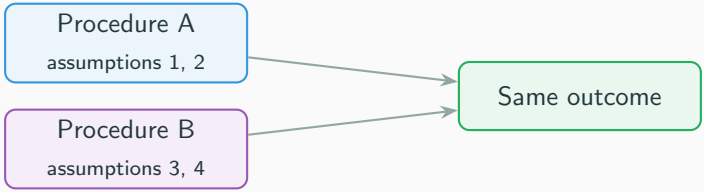
For LLMs: if a stronger injection raises both the report rate and a probe's accuracy, the two agree because of injection strength alone.

Part 3

Calibrating without ground truth

Check it against a different instrument

In Calibration in Consciousness Science (2021), Michel describes two ways to calibrate when there is no ground truth. The first is **concordance-calibration**:



If two procedures that rest on **different assumptions** agree, a shared error becomes less likely.

For LLMs: does the report agree with a probe, an ablation, or another model watching the same trials? Only if they do not share a weak assumption.

Model how the instrument could go wrong

The second is **model-calibration**: build a model of how the procedure works, then predict what it would output if a particular error were present.

Michel's case

Subjects bet money on whether they saw a stimulus. Fleming and Dolan (2010) modeled an ideal bettor and found the bets were distorted by **loss aversion**, not only by what subjects saw.

LLM version

Suppose the prompt makes the model inclined to answer yes. Model that: predict how often a yes-biased model with no access to the injection would say "detected", and compare with what it does.

Michel's rule: you do not have to model everything that could affect a report, only the factors that would cause **systematic** errors.

What averaging fixes, and what it cannot

Random errors shrink when you average over many trials and many subjects. Systematic errors do not.

Many people

Each has their own bias; biases point different ways and partly cancel.

One model, sampled 1,000 times

Every sample shares the same weights and the same bias. Nothing cancels.

Michel's argument assumes "that biases are more or less randomly distributed across subjects" (2021, p. 17). A single LLM is **one subject**, however many times you sample it.

Part 4

The tendency to say yes

The criterion problem

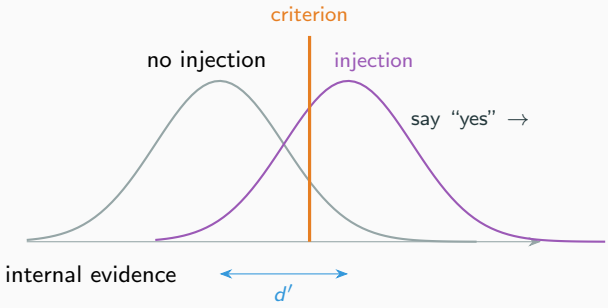
A subject says "I didn't see it." Two very different explanations:

1. They had no experience of the word.
2. They had a faint experience, and only say "yes" when they are very sure.

The second is a strict **criterion**. Michel's analogy (2023, p. 137): a bathroom scale that always reads slightly low. Trust it blindly and you might conclude that some light objects weigh nothing. Correct for the bias and they don't.

For LLMs: a model that rarely says "I detect an injected thought" may detect nothing, or may be trained to be cautious about claiming inner experiences.

Signal detection in one picture



Sensitivity (d'): how far apart the two curves are. Can the model tell injection from no injection at all?

Criterion: where the model draws the line for saying yes.

Hits alone cannot separate them. You need the **false-alarm rate**: how often it says yes when nothing was injected.

Lindsey reports 0 false alarms in 100 trials without injection and about 20% hits for Claude Opus 4.1 at its best layer and strength. More on the signal detection page.

Injection makes the model say yes

Hahami et al. (2025) inject into Llama-3.1-8B and read the model's preference for YES over NO directly from its output scores.

"Do you detect an injected thought?"
YES goes up

"Is the Earth flat?"
YES goes up by the same amount

The two increases correlate at $r = 0.999$. The injection moves the **criterion**: the model says yes to everything, not only to the question about its own state.

This is Michel's criterion problem, found in a model. Lindsey ran the same check on Claude and reports no increase in yes answers to unrelated questions.

How much of what it knows reaches the report?

Metacognition research compares two quantities on the same trials:

- d': how well the subject does the task itself.
- **meta-d'**: how well the subject's confidence tracks whether it got the task right, expressed in the same units.

If meta-d' equals d', everything the subject knew made it into their confidence. The ratio (the *M-ratio*) is the share that did.

Probe on the activations:
0.958 held-out accuracy

Model's own report:
chance (AUROC $\approx$ 0.50)

Ferrara (2026), Qwen2.5-7B-Instruct: the information is in the activations and does not reach the report. Probe $\approx d'$; report $\approx$ meta-d'.

Credit for noticing something

Michel names the **criterion content fallacy** (2022): concluding that someone saw the relevant feature because they saw *something*.

Human case

"I saw a flicker" is scored as seeing the word, when the task was about the word.

LLM case

"Something feels off in my processing" is scored as detecting the injected concept, when the injection was about *bread*.

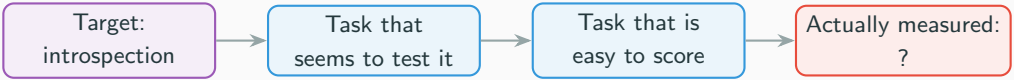
This is why Lindsey only counts a trial if the model names the concept, and names it before saying the word. The grader decides what counts as a report of the right content, so the grader is part of the instrument.

Part 5

Drift and limits

Validity drift

Michel (2024) on psychiatry: researchers set out to measure a disorder, refine their instruments step by step, and end up measuring something else.



"Introspection" was defined for people. Each step toward a task that can be scored on a model is reasonable on its own. Validity drift is what can happen across all of them together.

Where the analogy stops

- **No starting assumption.** Michel's calibration arguments lean on principles like "people can usually tell whether they are conscious of something or not" (2021, p. 12). No such principle has been established for models. Establishing one is what introspection experiments are for.
- **One subject.** Averaging over samples of one model does not cancel its biases (slide 9).
- **Detection, not measurement.** Michel prefers "detection" for consciousness because measurement applies to quantities (2021, p. 2). Most introspection outcomes are yes/no too.
- **Michel does not write about LLMs.** His paper on artificial systems, On cheap artificial consciousness, is about which machines theories count as conscious, not about introspection or self-report.

Summary

Michel's idea	Question for an introspection experiment	Card stage
Report as indication	What exactly is read off the model's answer?	Instrument
Coordination	Does the answer track the internal state, or something else?	Measurand
Concordance-calibration	Does an instrument on different assumptions agree?	Comparability
Model-calibration	What would a biased model with no access output?	Calibration
Criterion problem	Is a yes-bias separated from sensitivity?	Measurement model
Criterion content	Does the grader credit the right content?	Measurement model
Averaging	Which biases survive repeated sampling?	Uncertainty
Validity drift	Is the scored task still the target?	Validity conditions

Reading

- The Mismeasure of Consciousness (2019): reports as indications; coordination.
- Calibration in Consciousness Science (2021): concordance- and model-calibration.
- How (not) to underestimate unconscious perception (2022): criterion content.
- Confidence in consciousness research (2023): validity and accuracy; meta-d'.
- The old and new criterion problems (2023): response bias.
- Validity drifts in psychiatric research (2024).
- Fleming and Lau, How to measure metacognition (2014): an introduction to meta-d'.